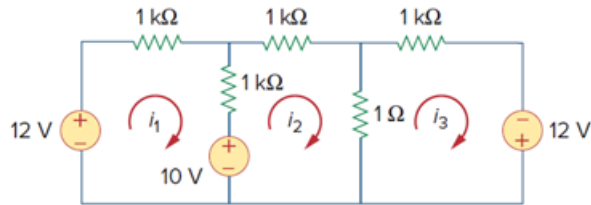


## Problem

Find the mesh currents  $i_1$ ,  $i_2$ , and  $i_3$  in the circuit in Fig. 3.99.

Figure 3.99:



## Step-by-step solution

## Step 1 of 5

Refer to Figure 3.99 in the textbook.

Apply mesh analysis in loop 1.

$$-12 + 1000i_1 + 1000(i_1 - i_2) + 10 = 0$$

$$-12 + 10 + 2000i_1 - 1000i_2 = 0$$

$$2000i_1 - 1000i_2 = 2 \quad \dots\dots (1)$$

Apply mesh analysis in loop 2.

$$10 = 1000(i_2 - i_1) + 1000(i_2) + 1(i_2 - i_3)$$

$$10 = 2001i_2 - 1000i_1 - i_3 \quad \dots\dots (2)$$

Apply mesh analysis in loop 3.

$$(i_3 - i_2) + 1000(i_3) = 12$$

$$1001i_3 - i_2 = 12 \quad \dots\dots (3)$$

## Step 2 of 5

Consider the matrix as,

$$\begin{bmatrix} 2000 & -1000 & 0 \\ -1000 & 2001 & -1 \\ 0 & -1 & 1001 \end{bmatrix} \quad \dots\dots (4)$$

Determine the  $\Delta$  value.

$$\begin{aligned} \Delta &= (2000((2001 \times 1001) - 1) + 1000(-1001 \times 1000)) \\ &= 3.005 \times 10^9 \end{aligned}$$

Determine the value of  $\Delta_1$ .

Consider the matrix for calculating  $\Delta_1$ .

$$\begin{bmatrix} 2 & -1000 & 0 \\ 10 & 2001 & -1 \\ -12 & -1 & 1001 \end{bmatrix}$$

$$\begin{aligned} \Delta_1 &= [2((1001 \times 2001) - 1) + 1000((10 \times 1001) - 12)] \\ &= 1.404 \times 10^7 \end{aligned}$$

**Step 3** of 5

Determine the value of  $\Delta_2$ .

Consider the matrix for calculating  $\Delta_2$ .

$$\begin{bmatrix} 2000 & 2 & 0 \\ -1000 & 10 & -1 \\ 0 & -12 & 1001 \end{bmatrix}$$

$$\begin{aligned} \Delta_2 &= [2000((10 \times 1001) - 12) - 2(-1000 \times 1001)] \\ &= 2.199 \times 10^7 \end{aligned}$$

Determine the value of  $\Delta_3$ .

Consider the matrix for calculating  $\Delta_3$ .

$$\begin{bmatrix} 2000 & -1000 & 2 \\ -1000 & 2001 & 10 \\ 0 & -1 & -12 \end{bmatrix}$$

$$\begin{aligned} \Delta_3 &= [2000((2001 \times -12) + 10) + 1000(12000)] \\ &= -3.6004 \times 10^7 \end{aligned}$$

**Step 4** of 5

Determine the current  $i_1$ .

Consider the equation for  $i_1$ .

$$i_1 = \frac{\Delta_1}{\Delta}$$

Substitute  $1.404 \times 10^7$  for  $\Delta_1$  and  $3.005 \times 10^9$  for  $\Delta$  in equation.

$$\begin{aligned} i_1 &= \frac{1.04 \times 10^7}{3.005 \times 10^9} \\ &= 3.46 \times 10^{-3} \\ &= 3.46 \text{ mA} \end{aligned}$$

Therefore, the current  $i_1$  is 3.46 mA.

**Step 5** of 5

Determine the current  $i_2$ .

Consider the equation for  $i_2$ .

$$i_2 = \frac{\Delta_2}{\Delta}$$

Substitute  $2.199 \times 10^7$  for  $\Delta_2$  and  $3.005 \times 10^9$  for  $\Delta$  in equation.

$$\begin{aligned} i_2 &= \frac{2.199 \times 10^7}{3.005 \times 10^9} \\ &= 7.31 \times 10^{-3} \\ &= 7.31 \text{ mA} \end{aligned}$$

Therefore, the current  $i_2$  is 7.31 mA.

Determine the current  $i_3$ .

Consider the equation for  $i_3$ .

$$i_3 = \frac{\Delta_3}{\Delta}$$

Substitute  $-3.6004 \times 10^7$  for  $\Delta_3$  and  $3.005 \times 10^9$  for  $\Delta$  in equation.

$$\begin{aligned} i_3 &= \frac{-3.6004 \times 10^7}{3.005 \times 10^9} \\ &= -0.012 \text{ A} \end{aligned}$$

Therefore, the current  $i_3$  is -0.012 A.